

**Title : Project Registration & Progress Review****FF No. 180**

|                                                                                                           |                              |
|-----------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Department: Computer Engineering</b>                                                                   | <b>Academic Year:2026-27</b> |
| <b>Semester :5</b>                                                                                        | <b>Group No. :TYG19</b>      |
| <b>Project Title: AgentShield: A Dynamic Risk-Aware Security Framework for the Model Context Protocol</b> |                              |
| <b>Project Area: AI And Cybersecurity</b>                                                                 |                              |

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Project approved / Not approved

**Guide****Project Coordinator****Head of Department**

**Project****Synopsis:****1. Introduction**

With the rapid adoption of Large Language Models (LLMs) and autonomous AI agents in enterprise environments, organizations are increasingly using the **Model Context Protocol (MCP)** to connect AI agents with internal tools, databases, and business applications. MCP provides a standardized communication framework that enables AI agents to interact with enterprise resources. However, granting direct access to enterprise systems introduces significant security challenges. AI agents may unintentionally perform unauthorized operations due to hallucinations, prompt injection attacks, or execution errors, potentially resulting in data breaches, unauthorized modifications, or disruption of critical business services.

Existing security solutions primarily focus on detecting vulnerabilities or securing individual MCP servers, but they often lack centralized runtime enforcement, dynamic risk evaluation, and comprehensive governance mechanisms. As organizations integrate multiple AI agents with enterprise systems, there is a growing need for an intelligent security framework capable of inspecting every interaction before it reaches critical infrastructure.

**AgentShield** is proposed as a dynamic security framework that operates between AI agents and MCP servers. Instead of allowing direct communication, the framework intercepts all MCP requests and responses, evaluates their associated risk, enforces configurable security policies, performs role-based authorization, supports Human-in-the-Loop (HITL) approval for sensitive operations, applies response sanitization to protect confidential information, and maintains detailed audit logs for accountability. By introducing a centralized security layer, AgentShield enables organizations to adopt AI technologies while ensuring security, governance, and regulatory compliance.

**2. Review of Literature****Paper 1: MCPShield – Dynamic Trust Evaluation for MCP Servers (2026)**

This research proposes a trust evaluation mechanism for MCP servers by analyzing server metadata, historical behavior, and runtime observations. The framework helps AI agents determine whether an external MCP server should be trusted before communication begins.

**Limitation:**

The approach focuses primarily on evaluating the trustworthiness of MCP servers and does not provide runtime policy enforcement, request interception, or enterprise-level access control mechanisms.

**Paper 2: MCPSec – Secure Communication Framework for Model Context Protocol (2026)**

MCPSec introduces protocol-level security improvements such as authentication, message integrity, cryptographic verification, and capability validation to strengthen secure communication between AI agents and MCP servers.

**Limitation:**

The framework secures the communication protocol itself but does not monitor or control the actions requested by AI agents during execution, nor does it provide dynamic authorization or approval workflows.

**Paper 3: ShieldMCP – Runtime Security Proxy for MCP (2026)**

ShieldMCP proposes a proxy-based architecture that intercepts communication between AI clients and MCP servers to detect suspicious requests and improve runtime security.

**Limitation:**

The framework mainly focuses on request interception and lacks integrated dynamic risk assessment, response-side data sanitization, centralized audit management, and flexible policy-driven authorization.

**Paper 4: Enterprise Security Framework for Model Context Protocol (2025)**

This paper discusses security governance for enterprise deployment of MCP-based AI systems by recommending identity management, policy enforcement, and organizational governance strategies.

**Limitation:**

The study provides conceptual guidelines but does not present a practical implementation capable of enforcing policies in real time.

**Paper 5: Security Threats and Attack Analysis for AI Agent Communication (2026)**

This research analyzes common threats including prompt injection attacks, excessive tool invocation, unauthorized access, information leakage, and denial-of-service attacks targeting AI agent infrastructures.

**Limitation:**

The research primarily focuses on threat identification and analysis without proposing a complete runtime framework capable of mitigating these attacks during execution.

**Research Gap**

From the literature survey, the following research gaps have been identified:

- Existing solutions primarily focus on detection rather than prevention of unauthorized AI actions.
- Most frameworks provide static policy enforcement without evaluating the contextual risk associated with each AI request.
- Human approval mechanisms for high-risk AI operations are largely absent.
- Response-side protection to prevent sensitive information leakage into LLM contexts is insufficient.
- Current approaches lack an integrated framework combining runtime interception, dynamic risk assessment, policy enforcement, data sanitization, rate limiting, and centralized monitoring.

AgentShield addresses these gaps by introducing a unified, risk-aware security framework specifically designed for enterprise MCP environments.

**3. Problem Statement**

The increasing integration of AI agents with enterprise systems through the Model Context Protocol has introduced significant security risks. AI agents may unintentionally execute unauthorized commands due to hallucinations, prompt injection attacks, excessive autonomous behavior, or improper access permissions. Existing security mechanisms either focus on protocol-level protection or post-execution monitoring, leaving enterprise systems vulnerable to runtime attacks and data leakage.

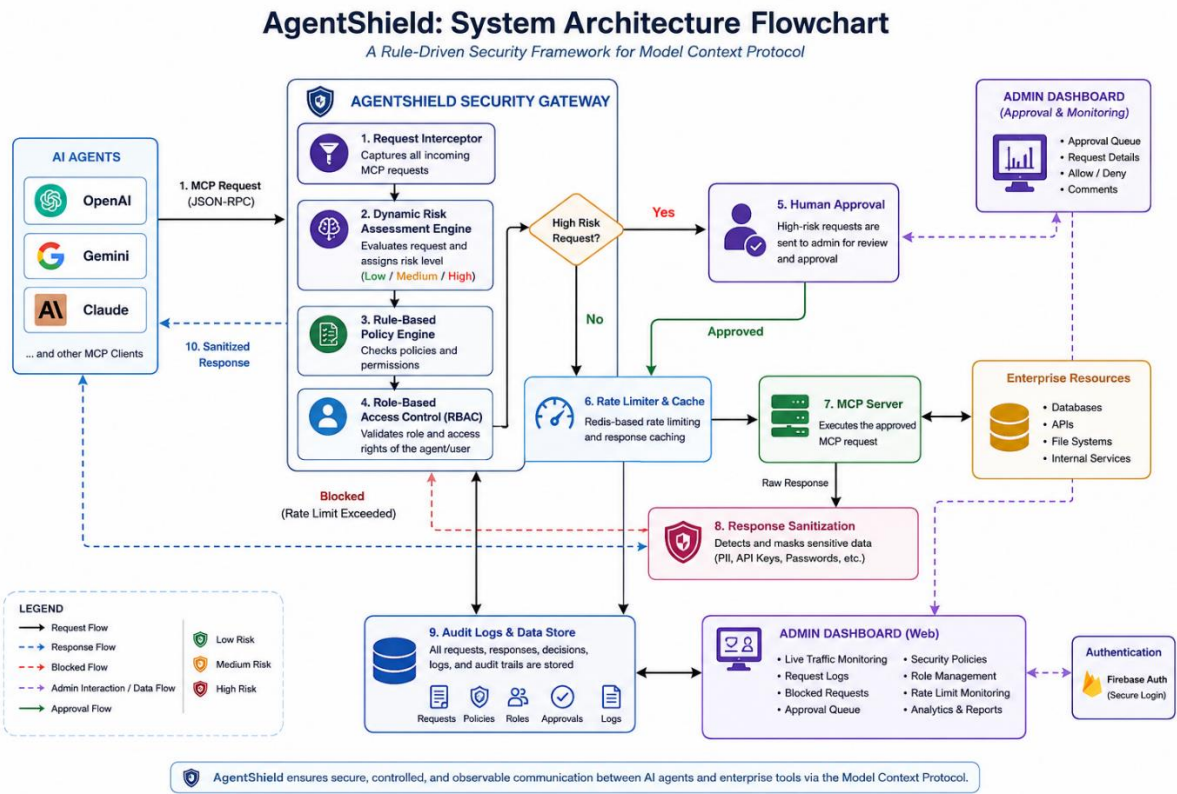
Therefore, there is a need for a centralized security framework capable of intercepting every MCP communication, evaluating the operational risk of AI requests, enforcing configurable security policies, preventing unauthorized operations, protecting sensitive enterprise information, and maintaining complete auditability without modifying existing MCP servers.

**4. Objectives**

The primary objectives of the proposed system are:

- To develop a centralized security framework for Model Context Protocol communication.
- To intercept and inspect all requests exchanged between AI agents and MCP servers.
- To implement a Dynamic Risk Assessment Engine that classifies requests into Low, Medium, and High risk categories.
- To enforce rule-based authorization policies according to user and agent roles.
- To implement Human-in-the-Loop approval for critical or high-risk operations.
- To perform response-side sanitization by masking Personally Identifiable Information (PII) before forwarding responses to AI agents.
- To implement Redis-based rate limiting for preventing excessive request generation and AI execution loops.
- To maintain comprehensive audit logs for monitoring, analysis, and compliance.
- To develop a web-based dashboard for policy management, request monitoring, and approval workflows.

5.SystemArchitecture



The proposed system follows a modular security gateway architecture positioned between AI agents and enterprise MCP servers.

Working Process

- AI agents (OpenAI, Gemini, Claude, etc.) generate MCP requests.
- AgentShield intercepts every incoming request.
- The Dynamic Risk Assessment Engine analyzes the request and assigns a risk level.
- The Rule Engine verifies whether the requesting agent possesses sufficient permissions.
- High-risk requests are paused and forwarded to the Human Approval module.
- Approved requests are subjected to Redis-based rate limiting before reaching the target MCP server.
- The MCP server executes the request and returns the response.
- The Response Sanitization module removes sensitive information such as API keys, passwords, credit card numbers, Aadhaar numbers, PAN numbers, and confidential identifiers.
- All request and response activities are stored in PostgreSQL for auditing.
- The React dashboard displays live traffic, policies, approval requests, audit logs, and blocked activities.

Proposed Technology Stack

| Component             | Technology                                    |
|-----------------------|-----------------------------------------------|
| Backend               | FastAPI (Python)                              |
| Frontend              | React                                         |
| Authentication        | Firestore Authentication                      |
| Database              | PostgreSQL                                    |
| Cache & Rate Limiting | Redis                                         |
| AI Clients            | OpenAI, Gemini, Claude                        |
| MCP Servers           | Dummy MCP Server, GitHub MCP, File System MCP |

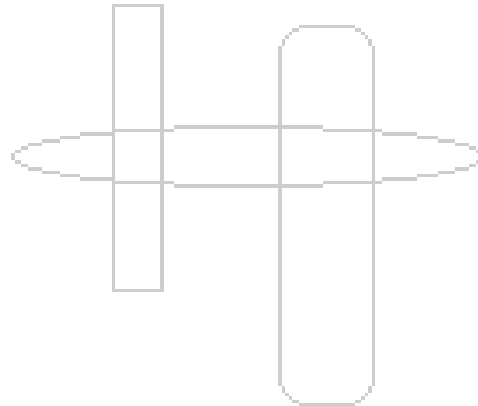
API Communication

JSON-RPC (Model Context Protocol)

**Expected Outcome**

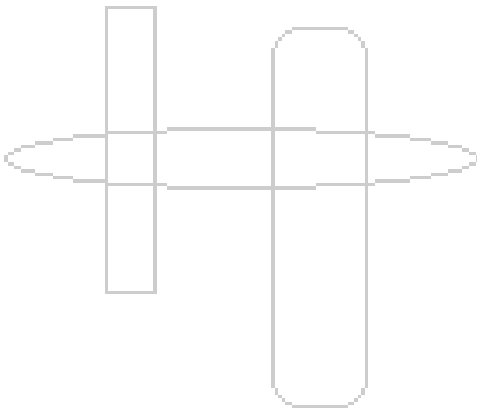
The proposed AgentShield framework will provide secure communication between AI agents and enterprise MCP servers by enforcing dynamic security policies, preventing unauthorized access, protecting sensitive organizational data, supporting administrator approval for critical operations, and maintaining centralized monitoring and audit capabilities. The framework is expected to improve the overall security, reliability, and governance of enterprise AI deployments while remaining compatible with existing MCP infrastructures.

**FF No 180**



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|-----------|-------------------|----------------------------------|--------------------|
| Group No. |                   |                                  |                    |
| Activity  | Review Schedule   | Progress Review Report submitted | Signature of Guide |
| Review 1  | Mid Sem. Semester | Yes / No                         |                    |
| Review 2  | End of Semester   | Yes / No                         |                    |

Format of Progress Review Report:

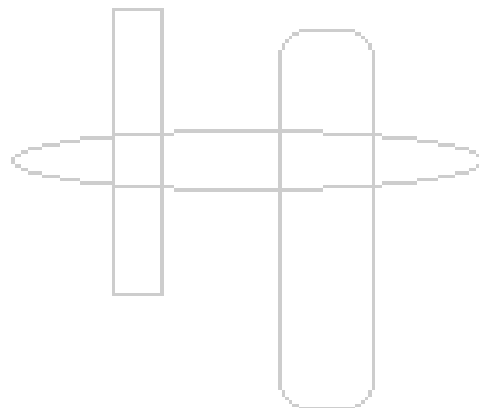
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| <b>Review No.: 1</b>                                                               | <b>Group No.:</b> | <b>Date:</b> |
| <b>Progress Review Report</b>                                                      |                   |              |
|  |                   |              |
| <b>Signature of Guide:</b>                                                         |                   |              |

**Review No.: 2**

**Group No.:**

**Date:**

**Progress Review Report**



**Signature of Guide:**